쉐도잉 앱 PRD 워크시트 — 영화 장면으로 만드는 쉐도잉 앱

인공지능과 영어교육 (ENG3088) · 쉐도잉 앱 프로젝트

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PRD — Movie-Clip Shadowing App

0. Movie Settings & Educational Use Notice (MUST appear in the app)

> **AI: Read this section first.** Every place in this PRD that says **[MOVIE TITLE]**, **[VIDEO ID]**, **[START]**, or **[END]** must use the values written here.

0-1. Movie Settings

Field	Value
Movie title (English) — **[MOVIE TITLE]**	Incredibles 2
Movie title (Korean)	인크레더블 2
Release year	2018
Director	Brad Bird
Screenplay	Brad Bird
Production company	Pixar Animation Studios
Distribution company	Walt Disney Studios
Scene description (한 줄, 예: 두 주인공의 첫 만남)	방에서 온 가족이 중식을 먹으며 슈퍼히어로 활동의 옳고 그름을 두고 언쟁을 벌이는 저녁 식사 장면
YouTube video title	Incredibles 2 (2018) - Dinner Scene (2/10)
YouTube channel name	Cartoon Clips
YouTube URL	https://www.youtube.com/watch?v=TbQ_yX_rxNI
Video ID (11 characters) — **[VIDEO ID]**	TbQ_yX_rxNI
Practice range start (seconds) — **[START]**	18
Practice range end (seconds) — **[END]**	67
Made by (내 이름)	정승교

0-2. Educational Use Notice

Educational use only / 교육 목적 전용

This app was made **only for non-commercial English learning** in the university course *AI and English Language Education* (ENG3088, Sungkyunkwan University). The film clip is used **only to practice listening and speaking (shadowing)**. It is not sold, shared for profit, downloaded, or edited. All rights to the film belong to its copyright holders.

이 앱은 성균관대학교「인공지능과 영어교육」(ENG3088) 수업의 **비영리 영어 학습 목적으로만** 제작되었습니다. 영화 장면은 듣기·말하기(쉐도잉) 연습에만 사용하며, 판매·영리적 배포·다운로드·편집을 하지 않습니다. 영화의 모든 권리는 저작권자에게 있습니다.

0-3. Where the app must show the notice and source

1. **Start screen:** a small card at the bottom titled "Educational Use & Source / 교육용 고지 및 출처". It shows the full notice (0-2) and these source lines from 0-1: movie title (English and Korean), release year, director, screenplay, production, distribution, scene description, YouTube video title, channel name, URL, and the practice range (written as m:ss – m:ss). The YouTube link opens in a new tab.
2. **Under the video player on every screen with a video:** one gray line (at least 13px): "Clip: *[MOVIE TITLE]* ([release year]), [distribution] / [production] · via YouTube · For educational use only"
3. **Finish screen:** the same one-line credit at the bottom.
4. Never remove or hide this notice. If a source field says "unknown", leave it out of the display.

1. App Name

🖍️ **Scene Shadow: Incredibles 2**

2. Why We Are Building This

Shadowing means listening to a native speaker and repeating the same sentence **almost at the same time**, copying rhythm, stress, and intonation. Movie scenes are great for this because the speech is natural and emotional. This app lets learners practice **short sentences** from one scene of the film **[MOVIE TITLE]**, one sentence at a time, with looping, slow speed, recording, and self-check.

🖍️ Why I chose this scene: The talks is slow simple and easy to undertand. The characters are also very distinct in their voices making distinguising them very easy.

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3. Target Users

- Korean high school 1st-year students (CEFR B1–B2).
- Used on a **mobile phone in portrait mode** (also works on a laptop).
- Session length: about 10–15 minutes.
- 🖍️ More about my learners: They know the vocabulary but are slow to come up with actual complete sentences they can speak as they have yet to see real examples of conversation.

4. Video Source — How to Bring the Movie Clip into the App (IMPORTANT)

Build the video part **exactly** like this:

- **Method:** Use the official **YouTube IFrame Player API** (`https://www.youtube.com/iframe_api`). Do **not** download, convert, or re-host the video. Do **not** use any third-party downloader.
- **Video ID:** use **[VIDEO ID]** from section 0-1.
- **Practice range of the whole app:** from **[START]** seconds to **[END]** seconds.
- **Player settings (playerVars):** `start: [START]`, `end: [END]`, `playsinline: 1`, `rel: 0`, `modestbranding: 1`, `controls: 1`, `cc_load_policy: 0`
- **Size:** 16:9, full width of the phone screen, placed at the top.
- **Playing one sentence (segment playback):**
  1. `player.seekTo(sentence.start, true)` then `player.playVideo()`.
  2. Check `player.getCurrentTime()` every 100 ms.
  3. When current time ≥ `sentence.end`, call `player.pauseVideo()`.
  4. If **Loop** is on, wait 1.5 seconds (time for the learner to speak), then play the same sentence again. If Loop is off, stay paused.
- **Speed:** use `player.setPlaybackRate()` with buttons **0.5× / 0.75× / 1.0×**.
- **Fallback:** If the video cannot be embedded (error code 101 or 150), show: "This video can't play inside the app. Tap to open it on YouTube." → opens `https://www.youtube.com/watch?v=[VIDEO ID]&t=[START]s` in a new tab. The sentence text, recording, and self-check must still work.
- The first play must start from a user tap (mobile browsers block autoplay with sound).

5. Screens (mobile portrait)

Screen 1 — Start

- App title (section 1) and a one-line explanation: "Listen. Copy. Speak at the same time."
- A short "What is shadowing?" card (3 bullet points, very simple English + Korean).
- Big button **Start Practice / 연습 시작**.
- Small link: **Edit Sentences / 문장 편집**.
- Educational Use & Source card (section 0-3).

Screen 2 — Sentence List

- The video player at the top.
- Practice sentences as cards: number, sentence text (can be hidden), speaker, and a progress mark (☆ not started, ◐ practiced, ★ done).
- Tap a card → Screen 3.

Screen 3 — Shadowing Practice (main screen)

1. Video player (16:9) with the one-line credit under it.
2. Sentence counter: "Sentence 3 / 8" with ◀ ▶ arrows.
3. Sentence text in large font (at least 22px), stress words in **bold**. A **Hide Text / 문장 숨기기** toggle.
4. Korean meaning (small, gray, can be hidden).
5. Control row: ▶ **Play**, 🔁 **Loop (on/off)**, speed **0.5× / 0.75× / 1.0×**.
6. **Step guide** (4 tabs, in order):
   - **Step 1 Listen** — text hidden, just listen 2–3 times.
   - **Step 2 Mumble** — text hidden, quietly follow the sound.
   - **Step 3 Read & Shadow** — text shown, speak together with the video.
   - **Step 4 Shadow** — text hidden, speak together with the video.
     Each tab shows one short instruction line in English + Korean.
7. **Record** area:
   - 🎙 **Record**: plays the sentence once and records the learner's voice at the same time. Recording stops 1 second after the sentence ends. (Loop is ignored while recording.)
   - ▶ **My Voice** to play the recording, and ▶ **Original** to replay the clip right after it.
8. **Self-check** (after recording), each with 😀 / 😐 / 😕 buttons:
   - "Did I keep up with the speed?"
   - "Did I stress the bold words?"
   - "Did my voice go up and down like the actor?"
9. **Word check (optional):** If the browser supports the Web Speech API, show **Check My Words**. Recognize speech in `en-US`, then color each word: matched words **green**, missed words **gray with underline**. Show "You said 7 of 9 words." If not supported, hide the button. Never show a failing grade.
10. **Next** ▶ button.

Screen 4 — Edit Sentences

- Video player and the current list.
- **Add Sentence** form: speaker, English text, Korean meaning, stress words, start time, end time.
- While the video plays, **Set Start = now** and **Set End = now** fill the time fields with `getCurrentTime()` (rounded to 0.1 s).
- **Test** plays just that segment.
- Rules: start and end must be between [START] and [END]; end greater than start; gentle warning if a sentence is longer than 4 seconds or 12 words.
- Edit / delete / reorder sentences.
- **Export** (copy JSON) and **Import** (paste JSON).

Screen 5 — Finish

- "Great job!" with the number of sentences completed and total recordings.
- A list of sentences marked 😐, with a **Practice again** button.
- One-line credit at the bottom.

6. Data Model

Store the sentence list and progress in `localStorage` (wrap in try/catch; if storage fails, keep data in memory). Each sentence:

```
{
  "id": 1,
  "speaker": "Character name",
  "text": "English sentence here",
  "korean": "한국어 뜻",
  "stress": ["word1", "word2"],
  "start": 24.0,
  "end": 26.5
}
```

Progress per sentence: `{ "id": 1, "plays": 0, "recordings": 0, "selfCheck": [null, null, null], "status": "new" }`. Status becomes "practiced" after one recording and "done" after a recording in Step 4.

7. Practice Sentences

Short lines (3–12 words) spoken between **[START] and [END]** seconds, written exactly as heard. Convert the filled rows into the JSON format in section 6 as the starting list. Ignore any row that still has 🖍️. If no rows are filled, start with an empty list and show: "No sentences yet. Tap Edit Sentences to add some."

#	Speaker	English sentence	Korean meaning	Stress words	Start (s)	End (s)
1	Dash	WHAT?! Is this all vegetables? Who ORDERED all vegetables?!	뭐라고요?! 이거 다 야채예요? 누가 야채만 다 시킨 거예요?!	WHAT, all, vegetables, ORDERED	19	23
2	Helen	I did. They're good and you're gonna have some.	내가 시켰어. 몸에 좋고 너도 좀 먹어야 해.	I, good, gonna, have	23	25
3	Violet	Are we going to talk about it?	우리 그것에 대해 이야기할 건가요?	talk, about	27	29
4	Violet	The elephant in the room.	모두가 알지만 말 안 하는 그 문제요.	elephant, room	30	32
5	Violet	I guess not then.	그럼 안 하겠네요.	guess, not	33	34
6	Helen	You're referring to today.	오늘 일 말하는 거구나.	referring, today	34	35
7	Dash	Yeah, what's the deal with today?	맞아요, 오늘 일은 대체 어떻게 된 거예요?	Yeah, deal, today	36	37
8	Helen	We all made mistakes. For example: you kids were supposed to watch Jack-Jack.	우리 모두 실수를 했어. 예를 들어, 너희들은 잭잭을 돌보기로 되어 있었잖아.	all, mistakes, example, supposed, watch, Jack-Jack	37	41
9	Violet	Babysitting, while you guys did the	애나 보라는 거였죠, 두 분이 중요한 일	Babysitting, you,	41	44

		important stuff.	을 아시는 중 안에요.	important		
10	Helen	We talked about this. You're not old enough to decide about these things.	우리 이 얘기 이미 했잖니. 너희는 이런 문제를 결정할 나이가 아니야.	talked, not, old enough, decide	45	48
11	Violet	We are old enough to help out.	우린 도울 만큼은 충분히 컸어요.	are, old enough, help out	48	50
12	Violet	Isn't that what you tell us, Dad?	아빠가 평소에 우리한테 그렇게 말씀하시잖아요?	tell, Dad	51	53
13	Bob	Yeah, well, "help out" can mean many different things.	그래, 음, '돕는다'는 건 여러 가지 다른 의미가 될 수 있지.	help out, mean, different things	53	56
14	Violet	But we're supposed to help, if there's trouble.	하지만 문제가 생기면 돕기로 되어 있잖아요.	supposed, help, trouble	56	59
15	Bob	Well, yeah, but...	글쎄, 그래, 하지만...	yeah, but	59	61
16	Violet	Aren't you glad we helped today?	우리가 오늘 도와서 기쁘지 않으세요?	glad, helped, today	61	63
17	Bob	Yeah, I know but...	그래, 나도 알지만...	know, but	63	65
18	Violet	You said that you were proud of us.	우리한테 자랑스럽다고 말씀하셨잖아요.	said, proud	65	67

8. Design Rules

- Mobile-first, portrait, one column, max width 480px centered on large screens.
- Main colors: Warm Yellow and Brown
- Buttons at least 48px tall, easy to tap with a thumb.
- Body text at least 18px; practice sentence at least 22px.
- All button labels in English with small Korean below (e.g., "Record / 녹음").
- Friendly tone. No scores that make students feel bad.

9. Technical Notes

- Single-page web app (React or plain HTML/JS), no login, no server, no paid API.
- Recording: `navigator.mediaDevices.getUserMedia({ audio: true })` + `MediaRecorder`. Keep only the latest recording per sentence in memory. Ask for microphone permission only when the learner first taps Record.
- If microphone permission is denied, show: "Mic is off. You can still listen and shadow out loud." and keep other features working.
- Speech recognition: `window.SpeechRecognition || window.webkitSpeechRecognition`, `lang = "en-US"` . Compare words after lowercasing and removing punctuation.
- Must work on Chrome (Android) and Safari (iPhone).
- Deployable to Vercel as a static site.

10. Done Checklist

- [ ] The YouTube video [VIDEO ID] loads with the official embed and plays only within [START]–[END].
- [ ] Play plays exactly one sentence and stops at its end time.
- [ ] Loop on repeats with a 1.5-second pause; Loop off stops after one play; speed buttons work.
- [ ] Hide Text and the 4 shadowing steps work.
- [ ] Record captures my voice while the clip plays; My Voice and Original play back.
- [ ] Self-check saves; Finish screen lists 😐 sentences.
- [ ] Edit mode: Set Start/End = now, Test, add/edit/delete, Export/Import JSON.
- [ ] Empty list shows a helpful message instead of an error.
- [ ] If embedding fails, the "Open on YouTube" fallback appears.
- [ ] "Educational use only" notice and full source appear on the Start screen; one-line credit under the video and on the Finish screen.